Phishing Email Detection & Awareness System

1. Introduction

Phishing is one of the most common cyber attacks targeting businesses and individuals. Attackers send fake emails pretending to be trusted organizations to steal sensitive information such as login credentials, financial details, or personal data.

This report analyzes a phishing email sample, identifies red flags, investigates the suspicious link, and provides awareness guidelines to help users avoid phishing attacks. The purpose is educational and for security awareness only.

2. Email Sample Analysis

Sample: Fax Notification Phishing Email

From: WirelessReceived attack@attacker.example.com

Subject: Monday, March 7 2022

To: Undisclosed recipients

Email Body Summary:

The email claims the user has received a new fax and asks them to click a link or attachment to view it.

Phishing Indicators Identified:

1. Suspicious Sender: 'attack@attacker.example.com' – Not a legitimate company domain.
2. Mass Email Targeting: "Undisclosed recipients" – Sent to multiple users.
3. Fake Notification: Pretends to be a SharePoint fax notification.
4. Urgency/Curiosity: Encourages user to act quickly by clicking the link.
5. Suspicious Link: 'https://attacker.example.com' – Not a verified company domain.

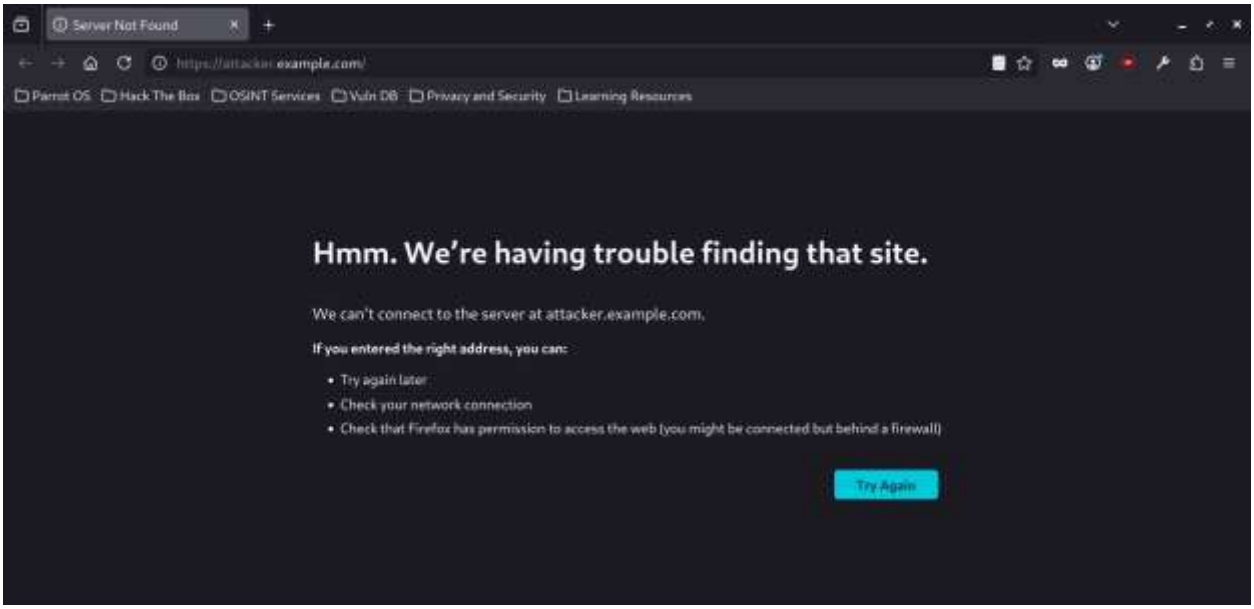
3. Risk Classification

Classification: PHISHING (High Risk)

Reason:

The email is designed to trick users into clicking a malicious link or downloading an attachment. The sender domain, generic recipients, and message structure all indicate a phishing attempt.

4. Phishing Link Investigation



Link Found in Email:

https://attacker.example.com

Observations:

- The link uses a suspicious domain designed to look legitimate.

- WHOIS query for the domain returned "No match", indicating it is a reserved/fake domain for educational purposes.
- In real attacks, similar links could lead to malware downloads or fake login pages.

Key Learning:

Even though this domain is fake, it demonstrates how attackers disguise phishing links to trick users. Always inspect links before clicking.

5. Domain Investigation

Domain: 'attacker.example.com'

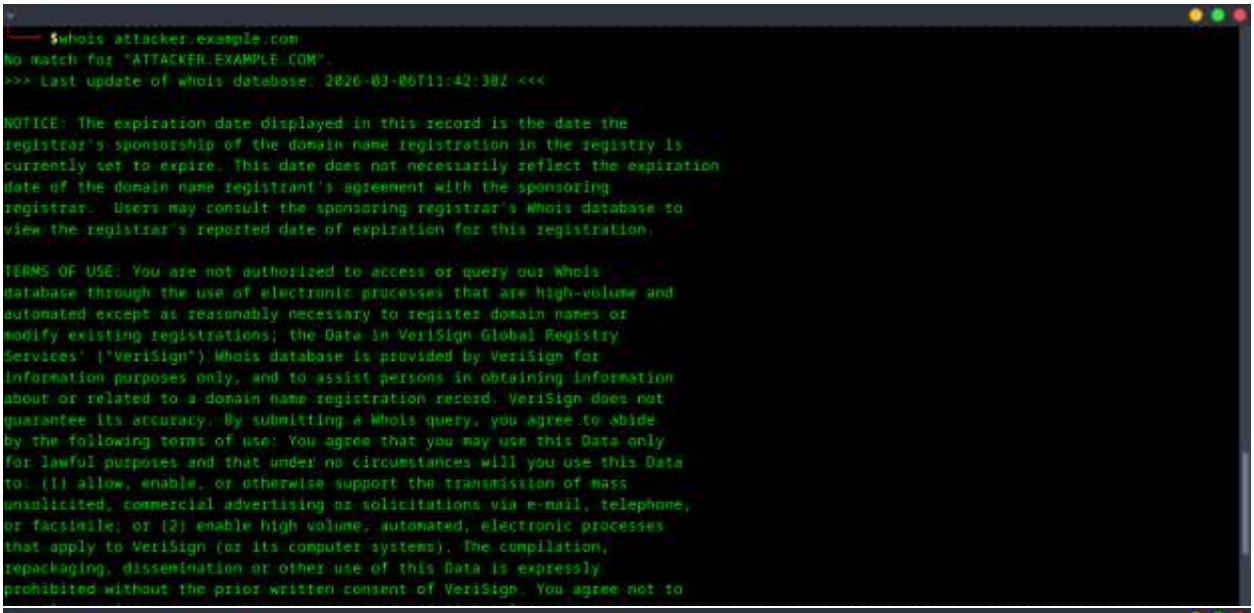
Tool Used: whois

Result:

=> WHOIS query returned "No match" because 'example.com' is a reserved domain for training and educational purposes.

=> Shows how attackers often use unregistered or recently registered domains for phishing campaigns.

Screenshots:



```
$whois attacker.example.com
No match for "ATTACKER.EXAMPLE.COM".
>>> Last update of whois database: 2026-03-06T11:42:38Z <<<

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
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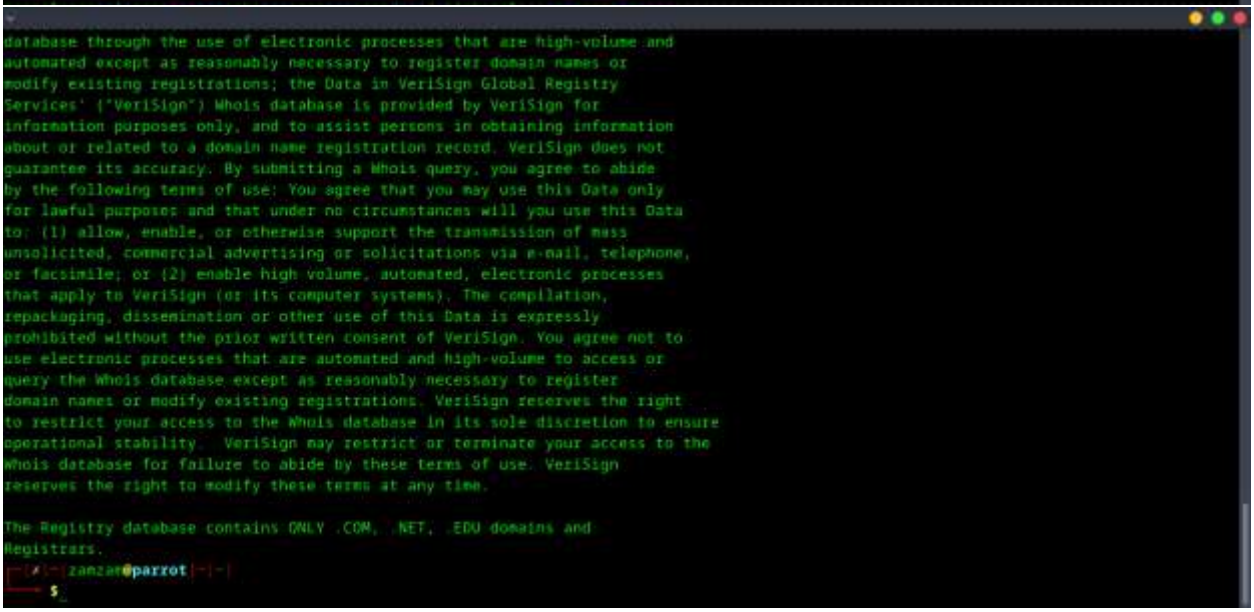

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to restrict your access to the Whois database in its sole discretion to ensure
operational stability.  VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
┌──(✗)─[~]─[zamzam@parrot]─[~]─
└──╼ $_
```

6.Phishing Prevention & Awareness Guidelines

Tips for Employees:

- Always check the sender’s email address.
- Hover over links before clicking to see the real URL.
- Verify unexpected attachments or links with IT/security team.

- Report suspicious emails immediately.
- Do not trust emails that create unnecessary urgency.

Do’s:

- Inspect email headers for suspicious domains.
- Keep security software updated.
- Verify links and attachments with official sources.

Don’ts:

- Do not click unknown attachments or links.
- Do not share passwords, OTPs, or credentials via email.
- Do not panic or rush actions due to fear-based messages.

7. Evidence

```
  GNU nano 7.2                                        message.eml
From: WirelessReceived WirelessReceivedWirelessReceived
 WirelessReceivedWirelessReceived WirelessReceivedWirelessReceived
 WirelessReceived <attack@attacker.example.com>
Subject: Monday, March 7 2022
Date: Mon, 7 Mar 2022 17:43:57 +0000
Message-ID: <CPWP152MB0PRGD.OUTLOOK.COM>
Content-Language: en-GB
Content-Type: multipart/alternative;
    boundary="_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_"
To: Undisclosed recipients:;
Return-Path: attack@attacker.example.com
MIME-Version: 1.0

--_000_CPWP152MB42210B2AAFBE0DF508B1251C9A089CPWP152MB4221LAMP_
Content-Type: text/plain; charset="Windows-1252"
Content-Transfer-Encoding: quoted-printable

SharePoint

You have a new fax! Click the attachment to view.

Print & Preview Here<https://attacker.example.com/>

                              [ line   1/291 ( 0%), col  1/56 (  1%), char     0/11581 ( 0%) ]
^G Help        ^O Read File   ^\ Replace     ^U Paste       ^G Go To Line  ^Y Redo        M-6 Copy       ^Q Where Was
^X Exit        ^R Where Is    ^K Cut         ^T Execute     ^_ Undo        M-A Set Mark   M-] To Bracket M-Q Previous
```

Server Not Found

https://attacker.example.com/

Parrot OS | Hack The Box | OSINT Services | Vuln DB | Privacy and Security | Learning Resources

Hmm. We're having trouble finding that site.

We can't connect to the server at attacker.example.com.

If you entered the right address, you can:

- Try again later
- Check your network connection
- Check that Firefox has permission to access the web (you might be connected but behind a firewall)

Try Again

```
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└──╼ $_
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